Wong Wei Jun

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EDUCATION

University of Wollongong — **Expected graduation year: 2028**

Bachelor of Science in Computer Science

SKILLS

Programming Languages: Python, Java, C++, JavaScript, TypeScript, HTML, CSS, SQL

Cloud & Infrastructure: AWS (ECS, EKS, Lambda, DynamoDB, S3, SNS), Terraform, Docker, GitHub Actions

Frameworks: FastAPI, Node.js, React

Developer Tools: Git, Linux, PostgreSQL, MongoDB, LocalStack, Prometheus, Grafana, Render, Neon, Vercel

Certifications: AWS Certified Solutions Architect – Associate, IBM Data Engineering, DeepLearning.AI Machine Learning Specialization

WORK EXPERIENCE

Tuition Teacher (Part-Time) | Incept Learning Centre, Desa Pandan — **Jan 2025 - July 2026**

- Taught IGCSE/SPM Mathematics, Sciences and English to students aged 7–16

Freelance Python Developer (Contract) | Coway New Era — **Oct 2025 - Feb 2026**

- Cut manual report generation time by over 90% (2 hours to under 10 minutes) by engineering a secure Python (Tkinter) automation tool that extracts data from the Coway Partner portal.
- Ensured uninterrupted, secure API/web access for repeated reporting runs by building session persistence with automated cookie retrieval and controlled manual login fallback.
- Enabled non-technical staff to pull accurate, custom-scoped customer reports without developer help by creating a dynamic filtering engine for date ranges and parameters, then exporting validated data to Excel.

PROJECTS

Driftwatch | ECS Fargate, DynamoDB, SNS, ECR, Terraform, CI/CD, Prometheus, Grafana | Demo Video

- Built drift detection to reduce AWS config-drift blind spots to ≤6 hours, reconciling live infrastructure against Terraform state across 5 resource types and paging high-severity diffs via SNS.
- Cut release lead time from ~20 min to <5 min by automating self-referential Terraform + GitHub Actions (OIDC → 147 pytest tests → Docker/ECR → ECS redeploy) on every merge
- Made API and drift-scan failures operable in production by adding Prometheus/Grafana dashboards and alert rules on scan failure rate and p95 duration

Cloud Job Runner Platform | EKS, Lambda, SQS, Fargate, ECR

- Built an event-driven job platform (API Gateway → Lambda → SQS → EKS Fargate) designed for ~100K jobs/day with durable delivery (delete-after-success + DLQ)
- Ensured zero message loss via SQS long-polling, delete-on-success pattern, Dead Letter Queue with 3-retry limit
- Reduced idle compute cost ~40% using KEDA, autoscaling worker pods (0–10) based on live queue depth.
- Added CloudWatch alarms and log metric filters to monitor queue depth and job failure rates in production

ShiftMate | FastAPI, React, PostgreSQL, Resend, Docker, Vercel, Render, Neon | Live Demo

- Eliminated account-creation friction by shipping a shift board with name/email signup, while giving coordinators token-based admin auth, real-time capacity enforcement, and CSV exports.
- Enabled one-click Google/Apple Calendar add for every signup by building a Resend + iCalendar confirmation pipeline that attaches .ics invites automatically on volunteer registration

EXPERIENCE IN COMPETITIONS

Gold Award, Malaysian Computing Challenge — **2024**

Finalist, National Secondary School Programming Contest (NSSPC) — **2024**